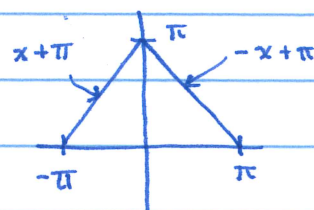


HW 13 Fourier Series

April 11, 2014

1. (i) $d_0 = \frac{1}{2\pi} (\pi^2) = \frac{\pi}{2}$

$$(n \neq 0) \quad d_n = \frac{1}{\pi n^2} (1 - (-1)^n) = \begin{cases} 0 & n \text{ even} \\ \frac{2}{\pi n^2} & n \text{ odd} \end{cases}$$

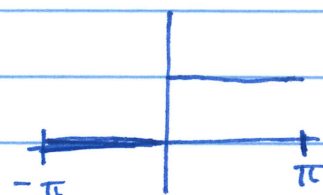


$$\therefore f(x) = \sum_{n=-\infty}^{\infty} d_n e^{inx} \quad \text{where } d_n = \begin{cases} \pi/2 & n=0 \\ \frac{1}{\pi n^2} (1 - (-1)^n) & n \neq 0 \end{cases}$$

$$\text{or } f(x) = \frac{\pi}{2} + \sum_{n \text{ odd}} \frac{2}{\pi n^2} e^{inx}$$

(ii) $d_0 = \frac{1}{2\pi} (\pi) = \frac{1}{2}$

$$(n \neq 0) \quad d_n = \frac{1}{2\pi i n} [1 - (-1)^n] = \begin{cases} 0 & n \text{ even} \\ \frac{1}{\pi i n} & n \text{ odd} \end{cases}$$

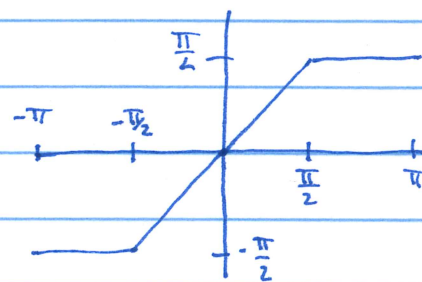


$$\therefore f(x) = \sum_{n=-\infty}^{\infty} d_n e^{inx} \quad \text{where } d_n = \begin{cases} \frac{1}{2} & n=0 \\ \frac{1}{2\pi i n} [1 - (-1)^n] & n \neq 0 \end{cases}$$

$$\text{OR } f(x) = \frac{1}{2} + \sum_{n \text{ odd}} \frac{1}{\pi i n} e^{inx}$$

(iii) $d_0 = 0$

$$(n \neq 0) \quad d_n = -\frac{1}{2in} (-1)^n + \frac{1}{2\pi n^2} ((-i)^n - i^n)$$



$$\therefore f(x) = \sum_{n=-\infty}^{\infty} d_n e^{inx} \quad \text{where } d_n \text{ is defined above.}$$

Area = 0